

with an export share of ₹2,400 crore (\$0.3 billion). The goal is to increase the domestic share to ₹26,400 crore (\$3.3 billion) and the exports to ₹88,000 crore (\$11 billion), with an envisioned investment of ₹17,600 crore (\$2.2 billion) in the next decade. India's recent achievements, including the Chandrayaan-3 lunar rover landing and the Aditya-L1 solar observatory mission, showcase its space exploration prowess. In 2022, Isro launched 36 OneWeb satellites and deployed the EOS-06 satellite along with eight nanosatellites. The sector is projected to reach \$13 billion by 2025, and India aims for close to 10 per cent share of the global space economy within a decade from now, potentially doubling the space industry's contribution to GDP and creating over 3 million jobs.

In India, achieving inclusive growth is closely tied to advancing digital infrastructure, service delivery, and literacy. Government initiatives like Digital India, Skill India, Make in India, Startup India, and Smart Cities are driving this transformation.

Satellite connectivity possesses the capability to optimise internet services in remote areas of India more effectively than terrestrial networks, overcoming limitations due to challenging topography or economic feasibility. The extension of internet services through satellite connectivity promises to promote educational equality, enhance social interactions, and strengthen public health, particularly in the face of pandemics or natural disasters.

Wider use of satellite data can revolutionise various sectors. In agriculture, farmers benefit by integrating satellite data to enhance yield. The energy sector uses it to monitor vegetation, prevent threats, and avoid power outages. The mining industry leverages satellite connectivity for productivity, emissions mapping, and supply chain monitoring, promoting sustainable practices. In insurance, improved

making it the world's second-most valuable start-up.

While the costs for heavy launches in low-earth orbit (LEO) have significantly decreased from \$65,000 per kg to under \$5,000 per kg, there is still much to achieve, including further reductions in launch and operational costs. Continued technology improvements are encouraging companies to increase their investments in the space economy.

The toughest problem to be solved in the next decade is to reduce the cost of launching satellites or humans into space. Strategies to achieve this goal include 3D printing, AI, and autonomous robotics to build a fleet of fully reusable, low-cost rockets, along with reusable components and stages.

In conclusion, driven by a burgeoning private sector, India should aim to significantly

reduce space exploration costs, paving the way for a space-based economy. The goal should be to shift from an earth-centric economic model to one that embraces the vast opportunities in space, as the current limitations on earth's resources can be overcome by making space access more affordable. **BT**

satellite imaging aids cost-effective risk assessment, particularly for global trade-related businesses. Recent advancements, coupled with reduced satellite costs, have led to innovations like higher-resolution sensors, enhancing image capture and data processing capabilities. Satellites now handle larger data than in the past.

The increasing importance the world gives to satellite technology has recently enabled SpaceX to achieve a valuation of \$180 billion,

making it the world's second-most valuable start-up.

The author is Co-founder and CEO of Skyroot Aerospace. Views are personal



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